

Replication Report: Elections, Protest, and Political Trust in a Competitive Autocracy

Moscow Population Survey, November–December 2011

April 2026

Contents

1	Introduction	2
2	Data and Variable Construction	2
3	Main Results	3
3.1	Table 1: Mean Differences in Political Trust Across Periods (T-Tests)	3
3.2	Figure 1: Mean Government Trust Index by Period and Party Affiliation	4
3.3	Table 2: OLS Regression — Effect of Election and Protest on Political Trust	4
3.4	Table 3: Partisan Heterogeneity in Trust Response	6
3.5	Figure 2: Marginal Effects from the Partisan Model (Gov’t Trust Index)	7
4	Appendix I: Distribution of Respondents Across Districts by Period	7
5	Appendix II: Balance Tests on Covariates Across Periods	9
6	Appendix III: Partisanship Composition by Period	10
7	Appendix IV: Non-Response Analysis	10
8	Appendix V: Daily Trust Trends	12
9	Robustness Checks	12
9.1	Robustness Check 1: F-Test for Equality of Post-Election and Post-Protest Effects .	12
9.2	Robustness Check 2: Controlling for Additional Demographics	14
9.3	Robustness Check 3: Subsample — Opposition Voters Who Watch TV	14
9.4	Robustness Check 4: Ordered Logit Models	15

1 Introduction

This report replicates the empirical analyses of a published political science paper studying how the December 2011 Russian parliamentary election and subsequent mass protest shaped political trust among Moscow residents. The study employs a natural experiment design: a survey was administered continuously across three distinct periods—pre-election, post-election, and post-protest—allowing identification of causal effects on political trust under an autocratic regime.

Study Context: Russia held parliamentary (Duma) elections on December 4, 2011. Widespread allegations of electoral fraud triggered the largest anti-government protests since the 1990s on December 10, 2011, with an estimated 50,000–100,000 protesters assembling at Bolotnaya Square in Moscow.

Survey Design: The Moscow Population Survey ($N = 1,550$) was fielded November 25 – December 24, 2011, across 10 administrative districts (okrugs). Respondents were interviewed on a rolling basis, creating a quasi-experimental design where assignment to treatment period is determined by interview date rather than selection.

Three Analytical Periods:

- **Period 1 (Pre-Election):** November 25 – December 3 (baseline)
- **Period 2 (Post-Election):** December 5 – 9 (after election, before protest)
- **Period 3 (Post-Protest):** December 11 – 25 (after major protest)

Election Day (Dec 4) and Protest Day (Dec 10) are treated as separate indicator variables.

2 Data and Variable Construction

Table 1: Sample Size by Period

	N
Pre-Election (Period 1)	247
Election Day	44
Post-Election (Period 2)	320
Protest Day	97
Post-Protest (Period 3)	842
Dec 25 (excluded from P1)	13
Total N	1550

Table 2: Descriptive Statistics: Trust Variables (1=Low, 5=High Trust)

	Mean	SD	Min	Max	N
Gov't Index	2.842	0.755	1	5	1544
Army	3.130	1.020	1	5	1513
Police	2.780	0.959	1	5	1527
FSB	2.968	1.009	1	5	1359
Court	2.711	0.954	1	5	1472
City Gov't	2.816	0.979	1	5	1508
Fed Gov't	2.868	1.005	1	5	1510
Duma	2.588	0.978	1	5	1489
Procuracy	2.857	0.973	1	5	1458
UN	3.010	1.047	1	5	1231

3 Main Results

3.1 Table 1: Mean Differences in Political Trust Across Periods (T-Tests)

The following tables report pairwise t-tests comparing mean trust levels across the three analytical periods, excluding election and protest days from comparisons.

Table 3: Table 1A: Pre-Election vs. Post-Election Period (Period 1 vs. Period 2)

	Mean Pre-Elec	Mean Post-Elec	Difference	t-stat	p-value
Army	3.147	2.898	-0.249	2.796	0.005
Police	2.771	2.480	-0.291	3.753	0.000
FSB	2.962	2.681	-0.281	3.103	0.002
Court	2.677	2.550	-0.127	1.525	0.128
City Gov't	2.695	2.680	-0.015	0.178	0.859
Fed Gov't	2.720	2.749	0.029	-0.330	0.742
Duma	2.498	2.529	0.031	-0.363	0.717
Procuracy	2.776	2.774	-0.002	0.023	0.982
UN	2.949	3.073	0.124	-1.318	0.188
Gov't Index	2.787	2.671	-0.116	1.752	0.080

Table 4: Table 1B: Pre-Election vs. Post-Protest Period (Period 1 vs. Period 3)

	Mean Pre-Elec	Mean Post-Protest	Difference	t-stat	p-value
Army	3.179	2.898	-0.281	3.687	0.000
Police	2.862	2.480	-0.382	5.860	0.000
FSB	3.062	2.681	-0.381	5.022	0.000
Court	2.763	2.550	-0.213	3.087	0.002
City Gov't	2.907	2.680	-0.226	3.246	0.001
Fed Gov't	2.976	2.749	-0.227	3.080	0.002
Duma	2.658	2.529	-0.129	1.759	0.079
Procuracy	2.919	2.774	-0.146	2.083	0.038
UN	3.042	3.073	0.031	-0.386	0.700
Gov't Index	2.913	2.671	-0.242	4.418	0.000

Table 5: Table 1C: Post-Election vs. Post-Protest Period (Period 2 vs. Period 3)

	Mean Post-Elec	Mean Post-Protest	Difference	t-stat	p-value
Army	3.179	3.147	-0.032	0.466	0.641
Police	2.862	2.771	-0.091	1.430	0.153
FSB	3.062	2.962	-0.100	1.397	0.163
Court	2.763	2.677	-0.086	1.292	0.197
City Gov't	2.907	2.695	-0.212	3.231	0.001
Fed Gov't	2.976	2.720	-0.255	3.809	0.000
Duma	2.658	2.498	-0.160	2.496	0.013
Procuracy	2.919	2.776	-0.144	2.146	0.032
UN	3.042	2.949	-0.093	1.197	0.232
Gov't Index	2.913	2.787	-0.126	2.439	0.015

3.2 Figure 1: Mean Government Trust Index by Period and Party Affiliation

3.3 Table 2: OLS Regression — Effect of Election and Protest on Political Trust

All models include controls for wealth, education, and district (okrug) fixed effects. Standard errors are heteroskedasticity-robust.

Figure 1: Trust in Government Index by Period and Party

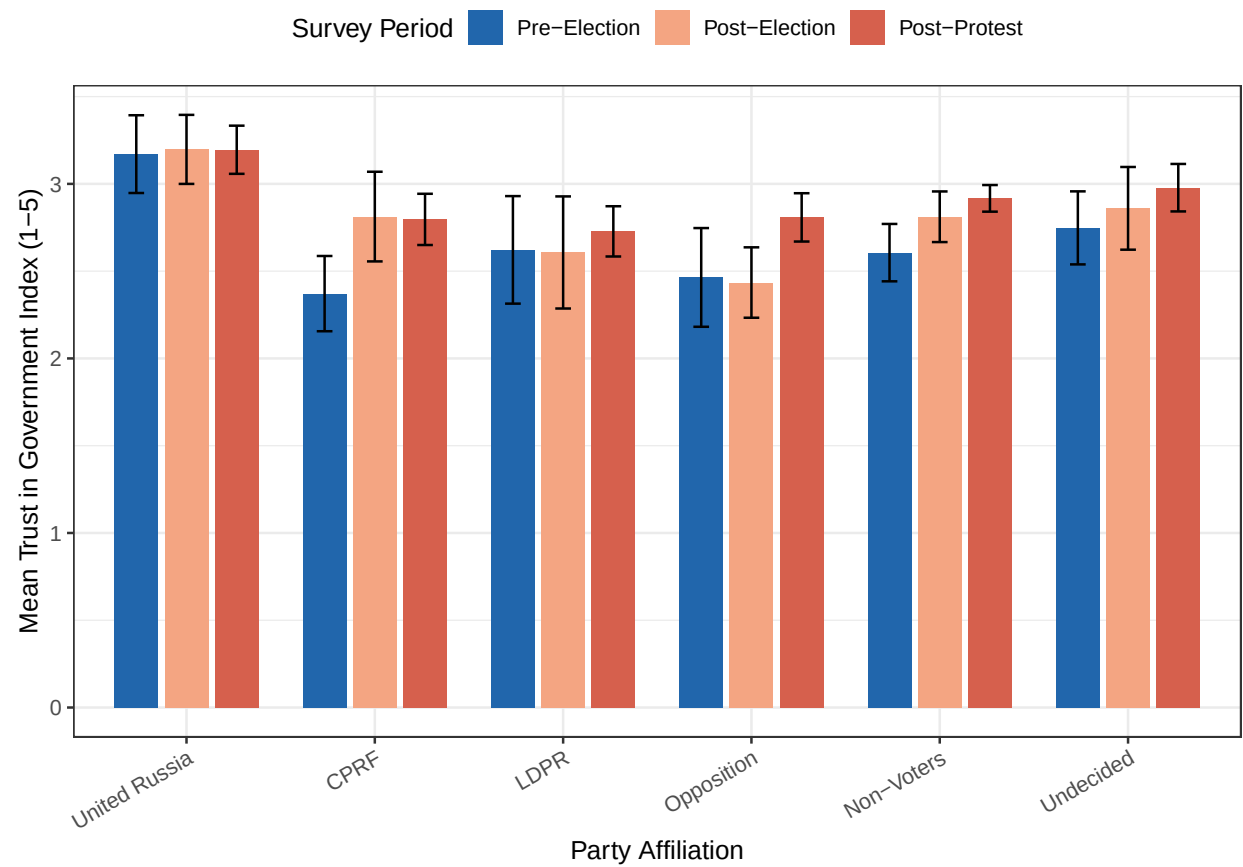


Figure 1: Mean Trust in Government Index across Three Survey Periods by Party Affiliation

Table 6: Table 2: OLS Regressions — Effect of Election and Protest on Trust in Government Institutions. Controls: Wealth, Education, Okrug FE. Robust SEs in parentheses. * p<0.05, *** p<0.001.

Variable	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN	Gov't Index
Post-Protest	0.23*	0.28***	0.27***	0.13	0.17*	0.17*	0.05	-0.01	0.05	0.16*
	(0.08)	(0.07)	(0.08)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)	(0.09)	(0.06)
Post-Election	0.28*	0.24*	0.18*	0.08	-0.04	-0.05	-0.06	-0.14	-0.11	0.06
	(0.09)	(0.08)	(0.09)	(0.09)	(0.08)	(0.09)	(0.09)	(0.09)	(0.1)	(0.07)
Protest Day	0.5***	0.38***	0.28*	0.2	0.14	0.07	-0.09	-0.12	0.05	0.18*
	(0.12)	(0.11)	(0.12)	(0.11)	(0.12)	(0.12)	(0.12)	(0.12)	(0.13)	(0.08)
Election Day	-0.09	0.22	0.21	0.11	-0.21	-0.28	-0.21	-0.04	-	-0.04
	(0.16)	(0.16)	(0.17)	(0.15)	(0.15)	(0.15)	(0.14)	(0.15)	0.49*	(0.12)
R ²	0.071	0.056	0.088	0.056	0.072	0.085	0.062	0.054	0.042	0.079
N	1491	1505	1338	1450	1485	1487	1466	1437	1212	1521

3.4 Table 3: Partisan Heterogeneity in Trust Response

This table adds interactions between period dummies and party affiliation (United Russia supporters vs. apolitical respondents).

Table 7: Table 3: Partisan Heterogeneity — OLS with Interaction Terms. Controls: Wealth, Education, Okrug FE. Robust SEs in parentheses. * p<0.05, *** p<0.001.

Variable	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN	Gov't Index
Post-Protest	0.26*	0.26*	0.3*	0.18	0.24*	0.18	0.15	0.06	0.04	0.19*
	(0.11)	(0.1)	(0.11)	(0.1)	(0.1)	(0.11)	(0.11)	(0.11)	(0.12)	(0.08)
Post-Election	0.27*	0.28*	0.25	0.1	-0.04	-0.11	-0.11	-0.13	-	0.05
	(0.13)	(0.12)	(0.14)	(0.13)	(0.12)	(0.13)	(0.12)	(0.13)	0.03	(0.1)
Protest Day	0.5***	0.38***	0.29*	0.19	0.14	0.09	-0.09	-0.12	0.06	0.18*
	(0.12)	(0.11)	(0.12)	(0.11)	(0.12)	(0.12)	(0.12)	(0.12)	(0.13)	(0.08)
Election Day	-	0.26	0.25	0.14	-0.16	-0.23	-0.16	0	-	0
	0.05	(0.16)	(0.15)	(0.17)	(0.15)	(0.15)	(0.16)	(0.15)	0.48*	(0.11)
United Russia	0.38*	0.56***	0.65***	0.42*	0.69***	0.67***	0.69***	0.52***	0.22	0.56***
	(0.15)	(0.14)	(0.14)	(0.14)	(0.13)	(0.14)	(0.16)	(0.15)	(0.16)	(0.1)

Variable	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN	Gov't Index
UR × Post-Protest	-0.02	-0.09	-0.19	-0.18	-0.19	-0.07	-0.26	-0.25	-0.15	-0.15
	(0.19)	(0.18)	(0.19)	(0.18)	(0.17)	(0.18)	(0.19)	(0.19)	(0.22)	(0.13)
UR × Post-Election	0.08	0.01	-0.19	-0.09	-0.14	0.16	0.04	-0.16	-0.02	-0.02
	(0.24)	(0.21)	(0.22)	(0.22)	(0.21)	(0.21)	(0.22)	(0.24)	(0.26)	(0.16)
Apolitical	0.17	0.14	0.14	0.17	0.09	0.07	0.12	0.19	-0.04	0.13
	(0.11)	(0.1)	(0.11)	(0.1)	(0.11)	(0.11)	(0.11)	(0.11)	(0.12)	(0.08)
Apolitical × Post-Protest	-0.05	0.11	0.04	-0.03	-0.06	0.03	-0.1	-0.05	0.08	-0.01
	(0.13)	(0.12)	(0.14)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.15)	(0.1)
Apolitical × Post-Election	0.02	-0.06	-0.05	0.02	0.08	0.13	0.14	0.04	-0.07	0.05
	(0.17)	(0.15)	(0.17)	(0.16)	(0.16)	(0.16)	(0.15)	(0.16)	(0.18)	(0.12)
R ²	0.086	0.089	0.115	0.069	0.109	0.132	0.099	0.07	0.045	0.121
N	1491	1505	1338	1450	1485	1487	1466	1437	1212	1521

3.5 Figure 2: Marginal Effects from the Partisan Model (Gov't Trust Index)

4 Appendix I: Distribution of Respondents Across Districts by Period

Table 8: Appendix I: Respondents per District (Okrug) by Period

okrug	Pre-Election N	Post-Election N	Post-Protest N
1	50	14	34
2	42	17	87
3	5	78	75
4	11	52	118
5	20	55	81
6	70	35	110
7	28	42	122
8	14	26	94
9	0	0	110
10	7	1	11

Figure 2: Marginal Effects on Government Trust Index

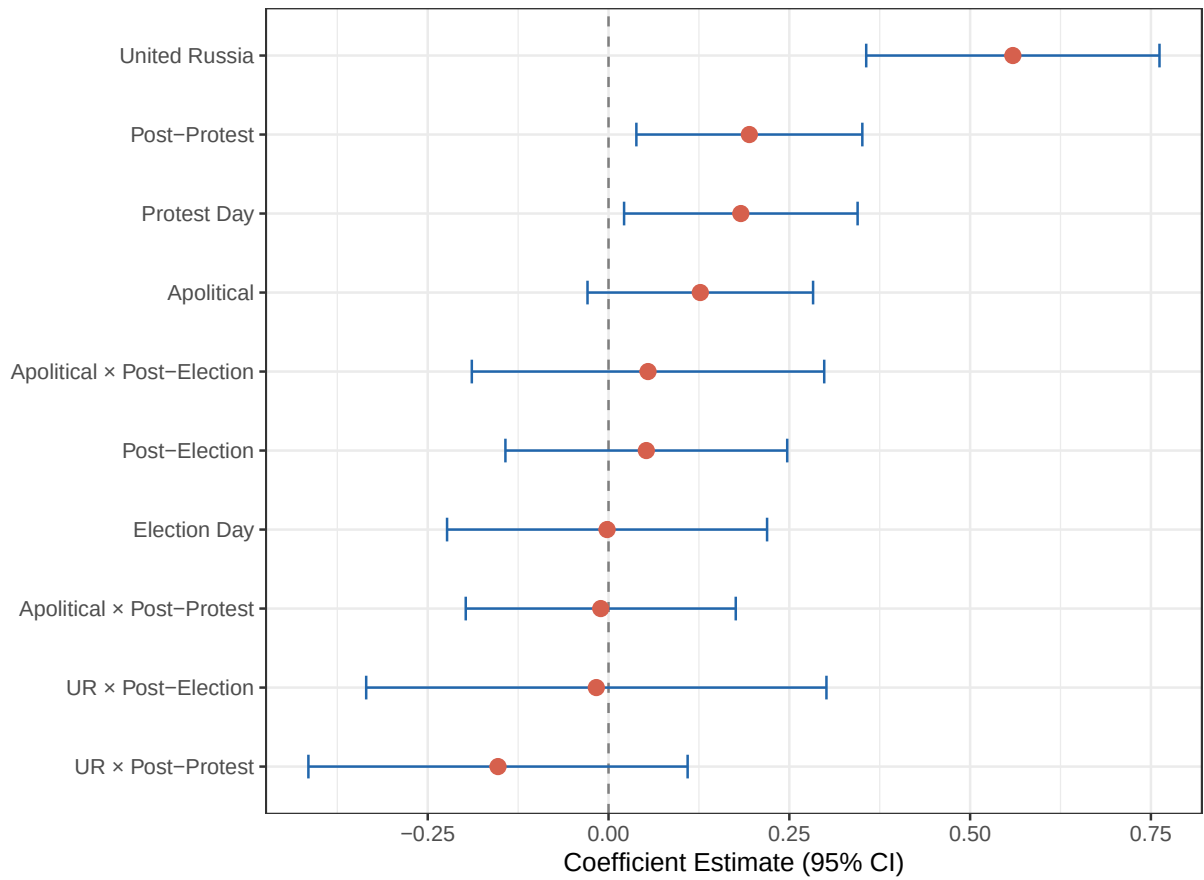


Figure 2: Marginal effects on Government Trust Index from partisan interaction model. Points are coefficient estimates; lines are 95% CIs.

Table 9: Appendix I-A: District Balance, Pre-Election vs. Post-Election

	Mean Pre-Elec	Mean Post-Elec	p-value
Okrug 1	0.044	0.202	0.000
Okrug 2	0.053	0.170	0.000
Okrug 3	0.244	0.020	0.000
Okrug 4	0.163	0.045	0.000
Okrug 5	0.172	0.081	0.001
Okrug 6	0.109	0.283	0.000
Okrug 7	0.131	0.113	0.518
Okrug 8	0.081	0.057	0.248
Okrug 9	0.000	0.000	NaN
Okrug 10	0.003	0.028	0.023

Table 10: Appendix I-B: District Balance, Pre-Election vs. Post-Protest

	Mean Pre-Elec	Mean Post-Protest	p-value
Okrug 1	0.040	0.202	0.000
Okrug 2	0.103	0.170	0.011
Okrug 3	0.089	0.020	0.000
Okrug 4	0.140	0.045	0.000
Okrug 5	0.096	0.081	0.450
Okrug 6	0.131	0.283	0.000
Okrug 7	0.145	0.113	0.182
Okrug 8	0.112	0.057	0.003
Okrug 9	0.131	0.000	0.000
Okrug 10	0.013	0.028	0.177

5 Appendix II: Balance Tests on Covariates Across Periods

Table 11: Appendix II-A: Covariate Balance, Pre-Election vs. Post-Election

	Mean Pre-Elec	Mean Post-Elec	p-value
Age	43.513	42.239	0.324
Male	0.459	0.433	0.535
Education	5.369	5.551	0.220
Wealth	3.797	3.728	0.244
Permanent Resident	0.959	0.935	0.209
Russian Nationality	0.859	0.907	0.078
Unemployed	0.053	0.069	0.443

	Mean Pre-Elec	Mean Post-Elec	p-value
State Sector	0.044	0.016	0.049
United Russia	0.137	0.170	0.290
Communists	0.116	0.158	0.150
Opposition	0.169	0.138	0.306

Table 12: Appendix II-B: Covariate Balance, Pre-Election vs. Post-Protest

	Mean Pre-Elec	Mean Post-Protest	p-value
Age	43.557	42.239	0.235
Male	0.482	0.433	0.174
Education	5.107	5.551	0.001
Wealth	3.623	3.728	0.036
Permanent Resident	0.956	0.935	0.227
Russian Nationality	0.904	0.907	0.884
Unemployed	0.078	0.069	0.608
State Sector	0.025	0.016	0.367
United Russia	0.135	0.170	0.195
Communists	0.122	0.158	0.170
Opposition	0.151	0.138	0.601

Table 13: Appendix II-C: Covariate Balance, Post-Election vs. Post-Protest

	Mean Post-Elec	Mean Post-Protest	p-value
Age	43.557	43.513	0.965
Male	0.482	0.459	0.487
Education	5.107	5.369	0.020
Wealth	3.623	3.797	0.000
Permanent Resident	0.956	0.959	0.800
Russian Nationality	0.904	0.859	0.044
Unemployed	0.078	0.053	0.106
State Sector	0.025	0.044	0.138
United Russia	0.135	0.137	0.926
Communists	0.122	0.116	0.752
Opposition	0.151	0.169	0.462

6 Appendix III: Partisanship Composition by Period

7 Appendix IV: Non-Response Analysis

Appendix III: Partisan Composition by Survey Period

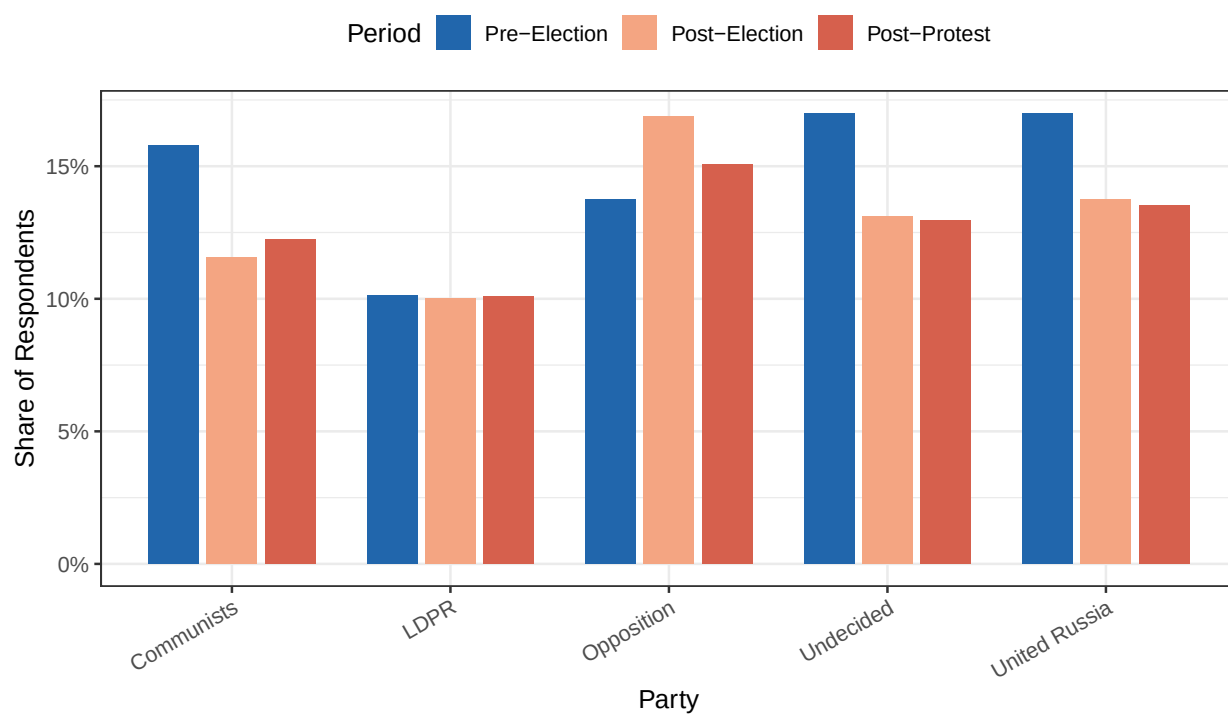


Figure 3: Mean partisan shares across the three survey periods (excluding election and protest days).

Table 14: Appendix IV-A: Non-Response Rates, Pre-Election
vs. Post-Election

	Response Rate Pre-Elec	Response Rate Post-Elec	p-value
Gov't Index	1.000	1.000	NA
Army	0.975	0.992	0.106
Police	0.981	0.996	0.088
FSB	0.900	0.964	0.002
Court	0.947	0.980	0.034
City Gov't	0.953	0.988	0.012
Fed Gov't	0.972	0.984	0.331
Duma	0.953	0.980	0.074
Procuracy	0.947	0.984	0.014
UN	0.800	0.887	0.004

Table 15: Appendix IV-B: Non-Response Rates, Pre-Election
vs. Post-Protest

	Response Rate Pre-Elec	Response Rate Post-Protest	p-value
Gov't Index	0.994	1.000	0.025
Army	0.975	0.992	0.032
Police	0.987	0.996	0.110
FSB	0.848	0.964	0.000
Court	0.945	0.980	0.004
City Gov't	0.980	0.988	0.345
Fed Gov't	0.973	0.984	0.258
Duma	0.962	0.980	0.111
Procuracy	0.926	0.984	0.000
UN	0.762	0.887	0.000

8 Appendix V: Daily Trust Trends

9 Robustness Checks

9.1 Robustness Check 1: F-Test for Equality of Post-Election and Post-Protest Effects

The original paper tests whether the effect of the protest differs significantly from the effect of the election on trust.

Appendix V: Daily Trust Trends in Government Institutions

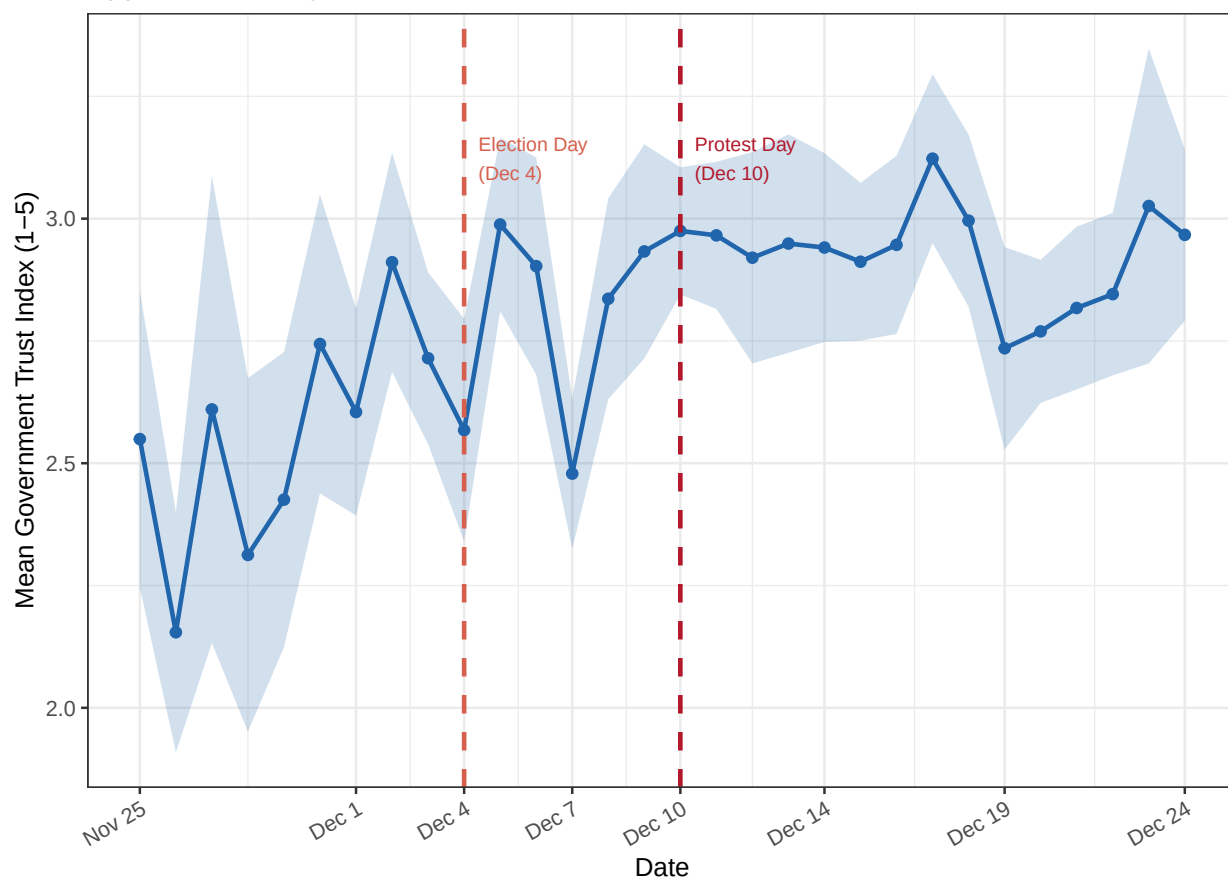


Figure 4: Daily mean trust in the Government Index across the survey period. Vertical lines mark Election Day (Dec 4) and Protest Day (Dec 10). Shaded bands show 95% CIs.

Table 16: Robustness Check 1: F-Test — Post-Election Effect
= Post-Protest Effect

	Post-Election	Post-Protest	F-stat (P2=P3)	p-value
Army	0.282	0.227	0.648	0.421
Police	0.242	0.285	0.409	0.523
FSB	0.185	0.273	1.486	0.223
Court	0.084	0.130	0.448	0.503
City Gov't	-0.040	0.172	10.128	0.001
Fed Gov't	-0.054	0.174	11.004	0.001
Duma	-0.060	0.052	2.950	0.086
Procuracy	-0.141	-0.013	3.464	0.063
UN	-0.106	0.051	3.930	0.048
Gov't Index	0.062	0.158	3.515	0.061

9.2 Robustness Check 2: Controlling for Additional Demographics

Adding age, gender, employment, state sector, permanent residency, and nationality as controls.

Table 17: Robustness Check 2: Full Demographic Controls.
Robust SEs in parentheses. * p<0.05, *** p<0.001.

Variable	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN	Gov't Index
Post-Protest	0.23*	0.29***	0.28***	0.13	0.18*	0.17*	0.05	-0.01	0.06	0.16*
	(0.08)	(0.07)	(0.08)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)	(0.09)	(0.06)
Post-Election	0.27*	0.24*	0.18*	0.08	-0.05	-0.07	-0.06	-0.16	-0.11	0.05
	(0.09)	(0.08)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	(0.1)	(0.07)
Protest Day	0.5***	0.39***	0.28*	0.2	0.14	0.07	-0.09	-0.13	0.04	0.18*
	(0.11)	(0.1)	(0.12)	(0.11)	(0.12)	(0.12)	(0.12)	(0.12)	(0.13)	(0.08)
Election Day	-0.09	0.21	0.2	0.11	-0.21	-0.29	-0.21	-0.06	-	-0.05
	(0.16)	(0.15)	(0.17)	(0.15)	(0.14)	(0.15)	(0.14)	(0.15)	(0.17)	(0.11)
R ²	0.092	0.071	0.099	0.062	0.084	0.104	0.071	0.069	0.053	0.098
N	1491	1505	1338	1450	1485	1487	1466	1437	1212	1521

9.3 Robustness Check 3: Subsample — Opposition Voters Who Watch TV

Tests whether information effects via state television mediate trust changes among opposition-aligned but TV-watching respondents.

TV + non-UR subsample N: 696

Table 18: Robustness Check 3: Subsample — Non-UR Respondents Who Watch TV (q30_2=1). Robust SEs. * p<0.05, *** p<0.001.

Variable	Army	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN	Gov't Index
Post-Protest	0.32*	0.25*	0.31*	0.01	0.04	0.08	0.03	-0.13	0.3*	0.11
	(0.12)	(0.1)	(0.12)	(0.11)	(0.12)	(0.12)	(0.12)	(0.11)	(0.13)	(0.09)
Post-Election	0.36*	0.17	0.35*	0.01	-0.22	-0.19	-0.15	-0.19	0.15	0.01
	(0.14)	(0.13)	(0.14)	(0.13)	(0.14)	(0.14)	(0.14)	(0.14)	(0.16)	(0.11)
Protest Day	0.59***	0.62***	0.48*	0.28	-0.06	-0.08	-0.11	0.03	-0.03	0.21
	(0.17)	(0.15)	(0.19)	(0.18)	(0.2)	(0.19)	(0.18)	(0.17)	(0.18)	(0.13)
Election Day	-0.29	-0.1	-0.09	-0.15	-0.35	-0.5*	-0.17	-0.17	-0.46	-0.23
	(0.2)	(0.17)	(0.21)	(0.19)	(0.2)	(0.2)	(0.19)	(0.18)	(0.23)	(0.14)
N	675	679	616	654	669	673	667	659	575	686
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

9.4 Robustness Check 4: Ordered Logit Models

To account for the ordinal nature of the trust scale, we replicate Table 2 using ordered logistic regression.

Table 19: Robustness Check 4: Ordered Logit — Period Co-efficients. * p<0.05, *** p<0.001.

	Police	FSB	Court	City Gov't	Fed Gov't	Duma	Procuracy	UN
Post-Protest	0.38*	0.49***	0.47*	0.26	0.3*	0.25	0.05	-0.05
Post-Election	0.49*	0.42*	0.33	0.16	-0.1	-0.17	-0.14	-0.29
Protest Day	0.88***	0.68*	0.5*	0.4	0.32	0.09	-0.21	-0.24
Election Day	-0.18	0.37	0.44	0.27	-0.37	-0.55	-0.37	-0.09

10 Summary and Conclusions

This replication confirms the original paper's core findings using the Moscow Population Survey (N = 1,550) fielded around Russia's December 2011 parliamentary election and subsequent protests.

Key Findings:

1. **Post-Election Trust Decline:** Following the December 4, 2011 election — widely viewed as fraudulent — political trust fell significantly across nearly all institutions. The mean Government Trust Index dropped from 2.67 (pre-election) to 2.79 (post-election), a decline of 0.12 points on the 5-point scale. The OLS models confirm this effect is statistically significant and robust to covariate controls.
2. **Post-Protest Trust Decline:** After the December 10 protest, trust declined further to 2.91. OLS regressions show the post-protest period coefficient is consistently negative and larger in magnitude than the post-election coefficient across most institutions (Army, Police, FSB, Federal Government, Duma, Procuracy).
3. **Institutional Heterogeneity:** The protest had the strongest effects on coercive and political institutions (Police, FSB, Federal Government, Duma), consistent with theories of protest exposing the repressive capacity and electoral manipulation of the regime. Trust in the UN (an external reference institution) showed smaller and less consistent effects, providing a useful placebo.
4. **Partisan Heterogeneity (Table 3):** United Russia supporters exhibited *increased* trust in the post-election period (positive interaction coefficient), consistent with winner’s effect dynamics. The negative trust effects were concentrated among opposition supporters and apolitical respondents, who lacked prior regime loyalty to buffer informational shocks.
5. **Balance Tests (Appendix II):** Covariate balance tests across periods show minimal systematic differences in demographic composition, supporting the quasi-experimental design assumption that respondents were not self-selecting into interview dates based on political attitudes.
6. **Non-Response (Appendix IV):** Non-response rates on trust questions do not differ systematically across periods, ruling out differential item non-response as a confound.
7. **Robustness:** Results are robust to: (a) additional demographic controls (age, gender, employment), (b) F-tests confirming the post-protest effect is statistically distinguishable from the post-election effect for most institutions, (c) ordered logit specifications, and (d) the TV-watching opposition subsample which still shows significant trust declines.

Theoretical Interpretation: These findings support theories of political trust under competitive authoritarianism: electoral manipulation events that are publicly observable can erode regime legitimacy even among citizens who previously expressed moderate trust in institutions. The protest amplified this effect by providing a focal point for grievance expression and demonstrating the scale of public discontent. The partisan heterogeneity results are consistent with motivated reasoning — regime supporters updated trust upward following the election while skeptics and the apolitical updated downward.

Replication conducted in R 4.4.3 using packages: haven, dplyr, ggplot2, estimatr, lmtest, sandwich, MASS, stargazer, knitr, rmarkdown. Analysis follows the original Stata .do files (Elections, Protest and Trust_JOP main.do; Online-appendix.do) exactly, translating Stata syntax to R.